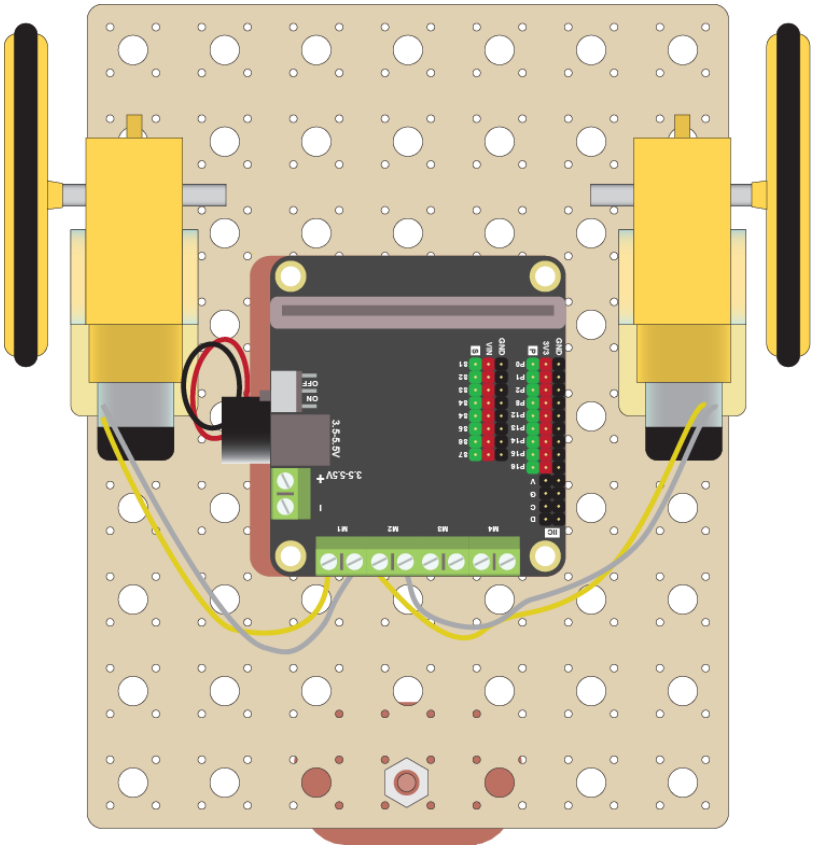
Build a Basic Robot

Project 1.01

In this workshop you will make a wheeled robot.



How it Works

The robot will have 3 wheels. Two large wheels will be driven by motors. The third wheel is a caster wheel, which is not driven, but will rotate in any direction that the other wheels demand.

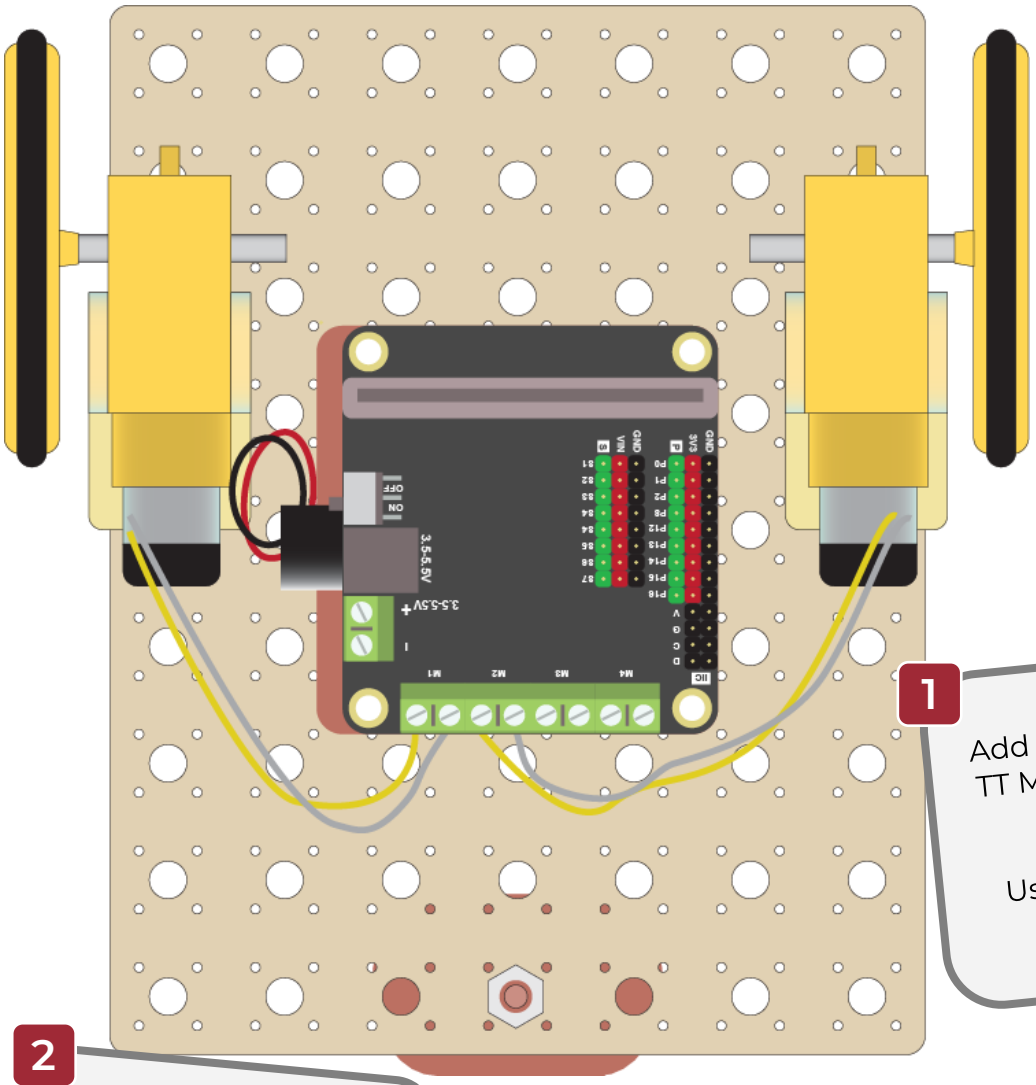
The Motor Controller is used to drive the motors. It turns on the current when you want the motor to spin and turns it off when you want the motor to stop. You can also change the speed and direction of the motor.

A Microbit will be coded to make whatever movements you want.

What to do

- Follow this worksheet to build and code the basic forwards movement of the robot.
- Attempt the challenges to get other movements—backwards, turning, dancing, or whatever you want!

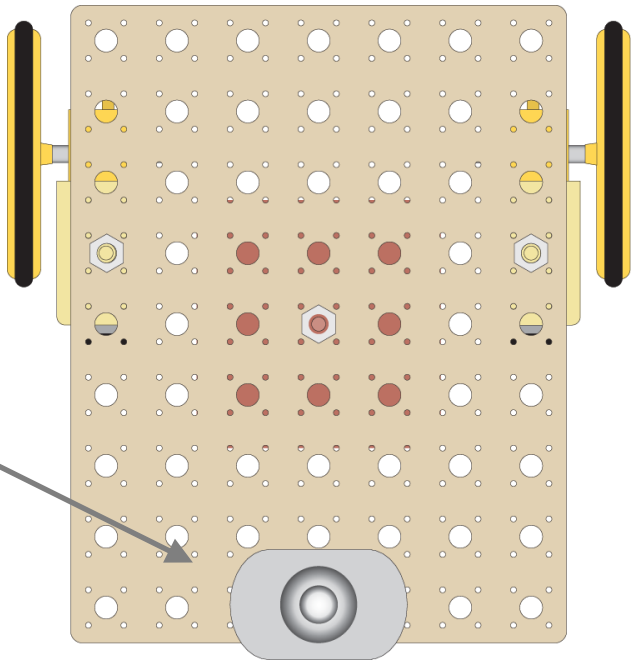
Assemble the Robot



1 Add the Motor Driver, TT Motors, TT Wheels and Battery.

Use nuts to secure them in place.

2 You don't have to use the exact layout shown here. Experiment with your own design!

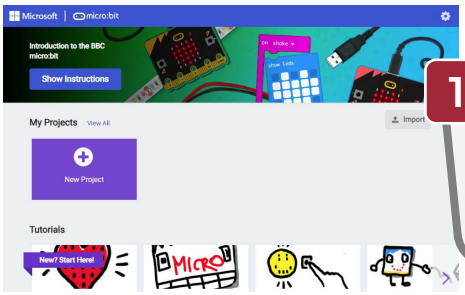


3 On the bottom, add the Caster Wheel

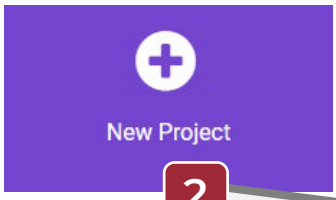
Build a Basic Robot: Code the Basic Movements

Create a Project for the Robot

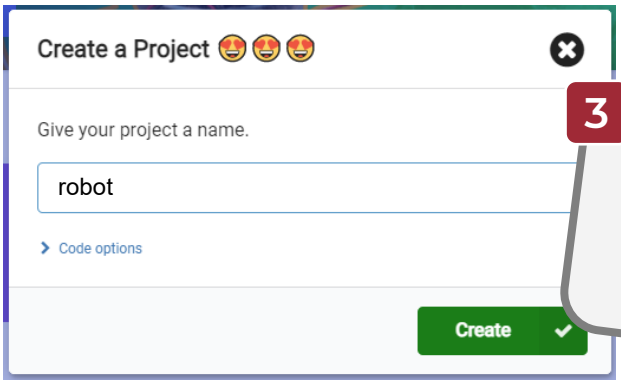
https://makecode.microbit.org/



1 Go to the Makecode website



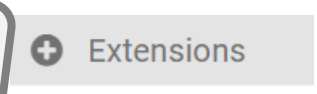
2 Click on New Project

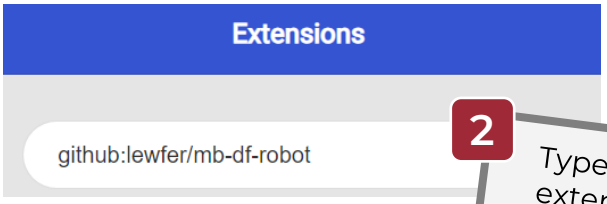


3 Give the project a name (whatever name you want!)

Add the Motor Driver Extension

1 Select this block

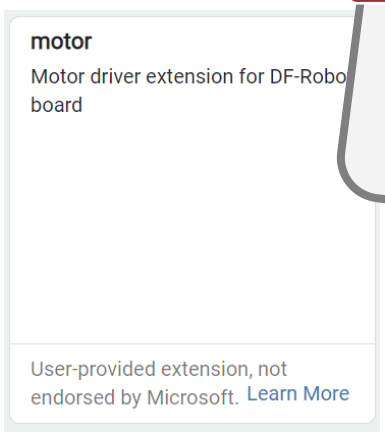




2 Type in the extension name

github:lewfer/mb-df-robot

3 Select the motor extension that appears



4 If all goes well you will see this new menu appear

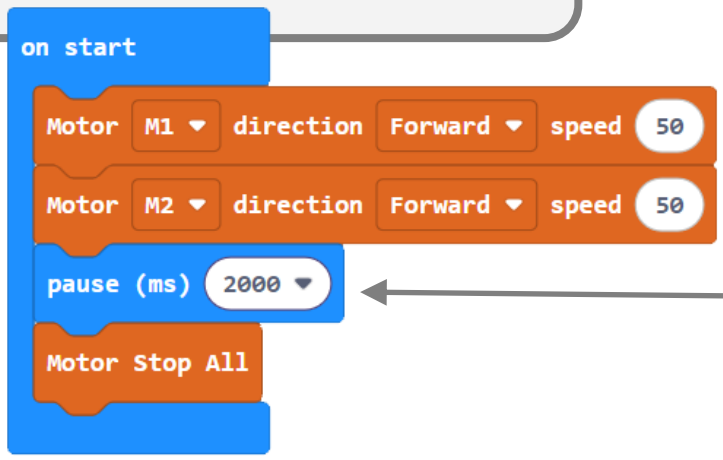


Add Code to Move the Motors

1 Find these code blocks

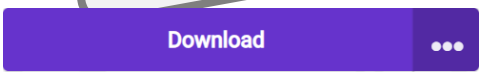


2 Add this code. This moves the robot forward at 50% speed for 2 seconds and then stops.

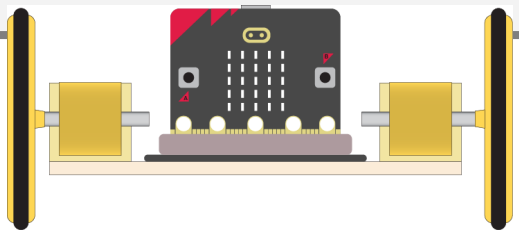


The pause tells the robot how long to carry on doing the previous thing. In this case this says how long the robot will keep moving forwards. The number is the time in milliseconds, There are 1000 milliseconds in 1 second.

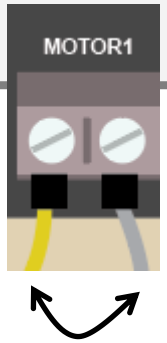
3 Download the code to the Microbit



4 Insert the Microbit into the Motor Controller board



5 Check that the robot runs correctly. If one or both of the wheels is turning the wrong way, swap the wires on the motor



Your challenge!

Get your robot to perform different movements:

- How can you get the robot to move backwards?
- How can you change the speed of the robot?
- How can you get the robot to turn?

Then try these:

- Get your robot to dance!
- Get your robot to make a perfect square!
- Design your own challenge for your robot!

Build a Basic Robot: Solutions

Moving Backwards

Use the “Reverse” direction to move backwards



Changing Speed

Click on the speed to adjust it



Turning

Adjust the speed of each motor to make a turn. Switch the speeds to turn the other way

Sharp Turn



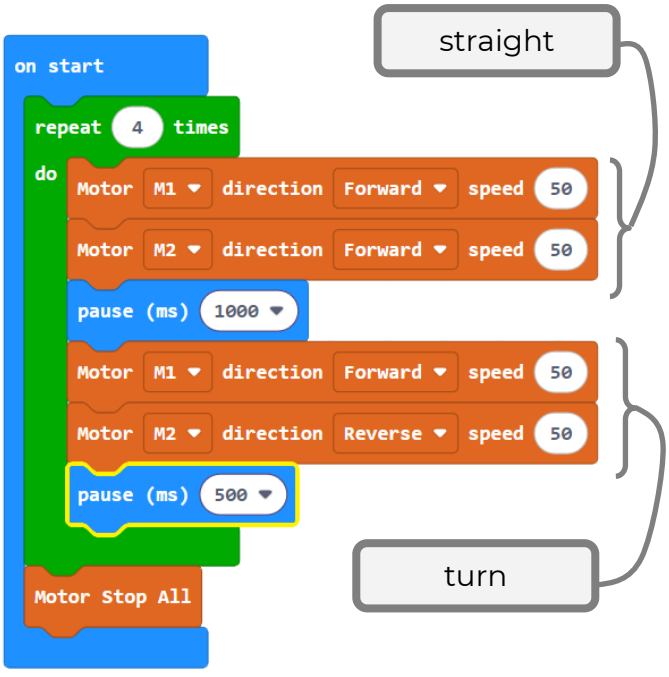
Gentle Turn



Spin



Square



Dance!

